Quantification: 536-63x_Hyp_RD_z15c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	9
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 9)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2685.89	2456	957.212	px^2
major_axis_length_px	lacuna shape	105.764	110.451	30.4809	px
minor_axis_length_px	lacuna shape	35.8255	33.4698	4.7954	px
aspect_ratio	lacuna shape	2.9651	3.2902	0.8493	unitless
eccentricity	lacuna shape	0.9202	0.9527	0.0606	unitless
solidity	lacuna shape	0.8089	0.794	0.0548	unitless
roots_count	headline	9.3333	9	4.3012	count
ring_length_r30_px	headline	316.778	319	98.1361	px
ring_length_r60_px	reported	827.222	916	255.137	px
owned_length_px	ownership-dependent	1548.93	1521.72	1239.9	px
edge_count	ownership-dependent	72.3333	56	59.3169	count
mean_edge_length_px	ownership-dependent	21.2532	20.8988	2.439	px
perimeter_px	normalised	316.102	326.25	94.662	px
ring_area_r30_px2	normalised	10655.1	10492	2115.79	px^2
ring_area_r60_px2	normalised	25893.6	26329	3766.38	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0295936	0.0304041	0.00618711	px^-1
ring_density_r60	normalised	0.0316773	0.0288124	0.00789022	px^-1
roots_per_100px_perimeter	normalised	2.9922	3.2276	1.0838	per 100 px
ring_attached_length_r30_px	network v2	278	249	113.358	px
ring_attached_length_r60_px	network v2	672.444	750	283.099	px
ring_length_w_r30_px	network v2	357.387	366.605	113.096	px
ring_length_w_r60_px	network v2	936.988	1036.16	289.854	px
sholl_crossings_r10	network v2	7.3333	7	2.7839	count
sholl_crossings_r20	network v2	10.7778	11	3.4197	count
sholl_crossings_r30	network v2	11.7778	12	3.0732	count
loop_count	loops, report only	5.7778	3	5.2863	count
loop_edge_share	loops, report only	0.2345	0.2308	0.146	share
ring_width_mean_r30_px	width	7.8132	8.051	0.7472	px

Per field
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measure	value	units
analysed_area_px2	1.02399e+06	px^2
total_skeleton_length_px	25028	px
canalicular_length_density_per_px	0.0244417	px^-1
junction_count	468	count
junction_density_per_px2	0.00045704	px^-2
lacuna_count	10	count

lacunae_per_px2	9.77e-06	px^-2
skeleton_component_count	350	count
mean_component_length_px	71.5086	px
median_component_length_px	19	px
field_length_density_w_per_px	0.027742	px^-1
field_density_without_flagged_per_px	0.0257413	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	607	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0244417	px^-1
field_length_density_with_unattached_per_px	0.0250344	px^-1
loop_count	59	count
loop_edge_share	0.1844	share
edge_count	1242	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7205	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	24982	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2274	76.8735	39.9156	1.9259	0.8546	0.8956	-0.3511	155.297	808.682	False
2	4398	148.873	40.64	3.6632	0.962	0.794	-0.278	209.417	635.271	False
3	2155	110.451	31.8217	3.4709	0.9576	0.7322	-0.4611	379.614	263.714	False
4	3362	116.899	43.0956	2.7125	0.9296	0.7535	-0.3179	479.939	367.637	False
5	414	41.7786	13.2327	3.1572	0.9485	0.9474	0.0032	469.732	4.9565	True
6	1029	48.7214	29.7491	1.6377	0.7919	0.8654	-0.7175	487.359	818.536	False
7	2767	117.614	33.0604	3.5575	0.9597	0.7731	-0.1563	645.45	527.102	False
8	3423	136.956	33.4698	4.0919	0.9697	0.856	-0.2992	677.154	397.595	False
9	2309	104.739	31.8331	3.2902	0.9527	0.7872	-0.2648	715.611	173.698	False
10	2456	90.7499	38.8445	2.3362	0.9038	0.8231	-0.397	792.997	804.486	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	179	480	170.853	9	18.9836	8.8036
2	16	394	960	3928.97	188	20.8988	7.7858
3	4	340	1067	1853.24	85	21.8028	8.2297
4	12	464	1214	2865.64	139	20.6161	8.3788
5	1	95	270	171.225	9	19.025	8.4072
6	6	222	529	641.014	34	18.8534	8.051
7	10	241	680	529.458	26	20.3638	7.3144
8	15	428	966	1855.46	87	21.3272	8.2971
9	7	319	916	1521.72	56	27.1736	6.9551

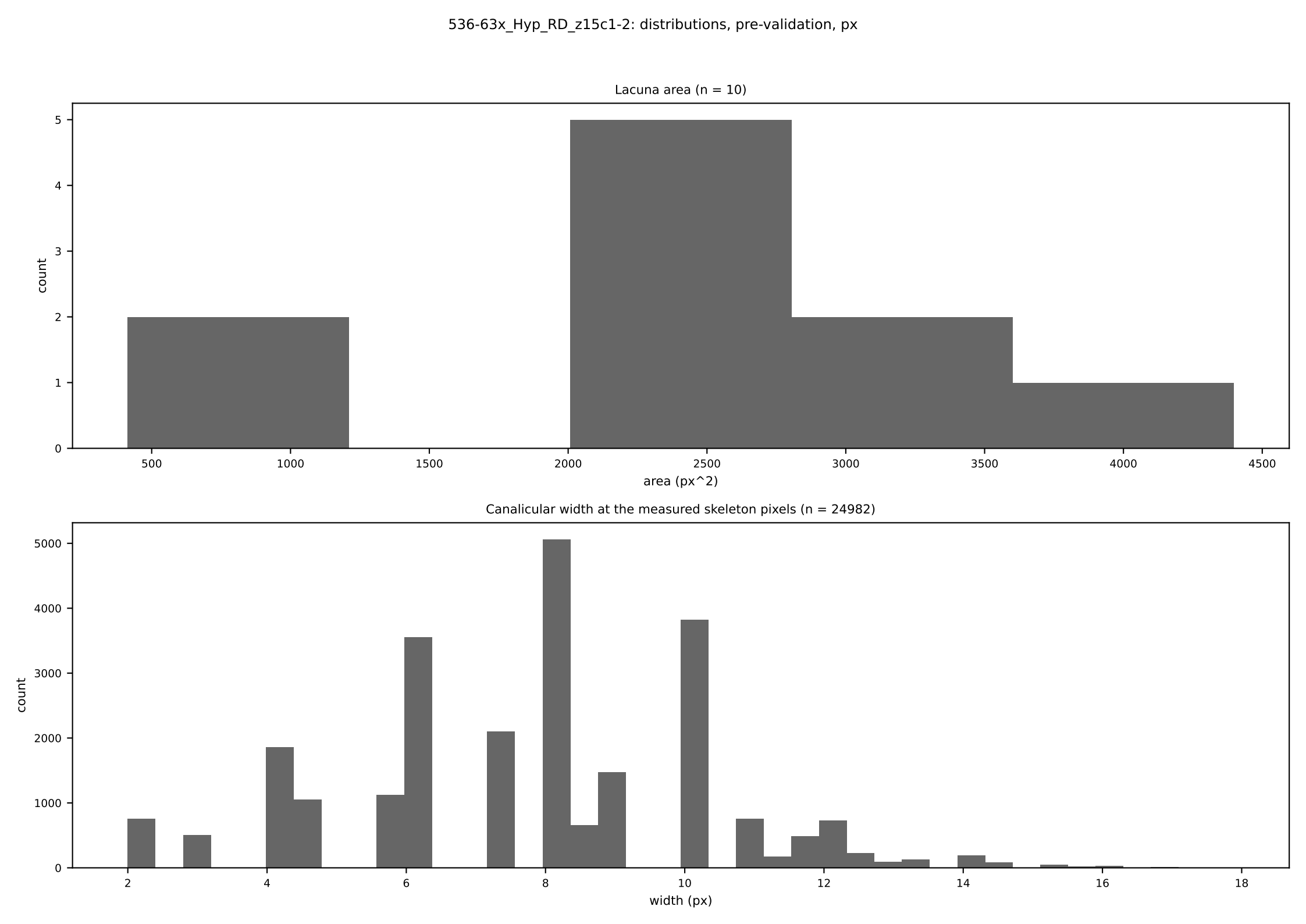
10 9 264 633 574.002 27 21.2593 6.5034

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	258.593	8854	23227	1	1	0.0202169	0.0206656	1.9335
2	480.115	14058	33319	1	1	0.0280267	0.0288124	3.3325
3	293.363	10490	25563	1	1	0.0324118	0.04174	1.3635
4	371.789	12101	27675	1	1	0.0383439	0.0438663	3.2276
5	94.5269	3043	8912	0.5484	0.5321	0.0312192	0.0302962	1.0579
6	142.61	6775	19178	1	1	0.0327675	0.0275837	4.2073
7	364.475	11610	26508	1	1	0.020758	0.0256526	2.7437
8	326.25	11953	26656	1	1	0.0358069	0.0362395	4.5977
9	353.961	10492	26329	1	1	0.0304041	0.0347905	1.9776
10	253.764	9563	24587	1	1	0.0276064	0.0257453	3.5466

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	103	297	194.953	537.889	2	6	7
2	371	866	440.019	1090.03	10	12	15
3	315	860	386.291	1193.38	9	13	13
4	435	1125	535.831	1390.86	11	15	16
5	69	164	104.77	304.35	1	2	4
6	203	500	251.965	602.044	6	7	11
7	198	420	271.936	796.293	5	7	8
8	419	859	475.191	1082.88	9	15	12
9	249	750	366.605	1036.16	7	11	14
10	209	375	293.693	703.357	7	11	10


536-63x_Hyp_RD_z15c1-2: distributions, pre-validation, px
Lacuna area (n = 10)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 24982)
count
width (px)